$$W = 1 - \left(\frac{e\left(1 + \frac{1}{n}\ln(1-\alpha) + o\left(\frac{1}{n}\right)\right) - 1}{e-1} \right)^n =$$

$$= 1 - \left(\frac{(e-1) + \frac{e}{n}\ln(1-\alpha) + o\left(\frac{1}{n}\right)}{(e-1)} \right)^n =$$

$$= 1 - \left(1 + \frac{1}{n}\,\frac{e \cdot \ln(1-\alpha)}{e-1} + o\left(\frac{1}{n}\right)\right)^n =$$

$$= 1 - e^{\frac{e \cdot \ln(1-\alpha)}{e-1}} = 1 - e^{\ln(1-\alpha)^{\frac{e}{e-1}}} =$$

$$= 1 - (1-\alpha)^{\frac{e}{e-1}} \xrightarrow[n \to \infty]{} 1$$

$\Rightarrow$ Критерий не состоятелен

№ 12

$n = 2 \qquad \alpha = 0{,}2$

H_0:

грань	1	2	3	4
p	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{3}$

H_1:

грань	1	2	3	4
p	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

$$\ell = \frac{L_1}{L_0} = \frac{\left(\frac{1}{4}\right)^2}{\left(\frac{1}{4}\right)^{n_1} \cdot \left(\frac{1}{4}\right)^{n_2} \cdot \left(\frac{1}{3}\right)^{n_3} \cdot \left(\frac{1}{6}\right)^{2-n_1-n_2-n_3}}$$

uczog	ℓ	p
(3,3)	2.25	0.0278
(3,1), (1,3)	1.5	0.042
(3,2), (2,3)	1.5	0.042
(4,3), (3,4)	1.125	0.056
(2,2), (1,2), (1,1)	1	0.0625

$\Sigma p = 0.1958$

$P(\ell \geq c \mid H_0) = \alpha$

$$G = \{(3,3), (3,1), (1,3), (2,3), (3,2)\}$$

$$W = P(\ell \geq c \mid H_1) = 5 \cdot \left(\frac{1}{4}\right)^2 = 0{,}3125$$